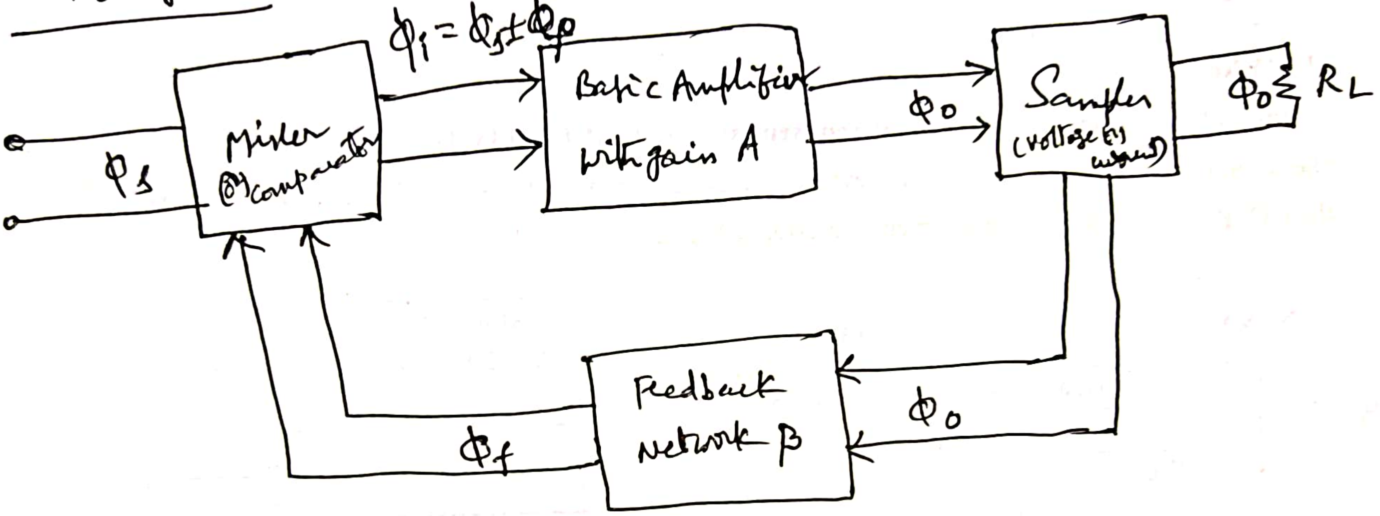
3.0 Feedback Amplifiers

Basic concept of feedback:-

A portion of o/p signal is taken from the o/p of the amplifier and is combined with normal i/p signal and there by feedback is accomplished.

Block diagram



The o/p quantity (either voltage or current) is sampled by a suitable sampler, ~~namely~~, voltage sampler or current sampler and fed to feedback network. The o/p of feedback N/W (signal) has a fraction of o/p signal combined with external source signal ϕ_s through a mixer and to the basic amplifier. Mixer also known as comparator is of two types, series mixer and shunt mixer.

A - gain of basic Amplifier $= \frac{\phi_o}{\phi_i}$, β = feedback ratio $= \frac{\phi_f}{\phi_o}$

A_f = gain of feedback amplifier $= \frac{\phi_o}{\phi_s}$

ϕ_s = ac signal in the i/p side

ϕ_f = feedback signal

Positive feedback :-

If ϕ_f is inphase with ϕ_s, then the effect of feedback will increase the i/p signal given to amplifier, then $\phi_i = \phi_s + \phi_f$. Hence, the i/p voltage applied to the basic amplifier is increased thereby increasing ϕ_o exponentially. This type of feedback is said to be positive (or) regenerative feedback.

Gain of Amplifier with +ve feedback is,

$$A_f = \frac{\phi_o}{\phi_s} = \frac{\phi_o}{\phi_i - \phi_f} = \frac{1}{\frac{\phi_i}{\phi_o} - \frac{\phi_f}{\phi_o}}$$

$$= \frac{1}{\frac{1}{A} - \beta} = \frac{A}{1 - A\beta}$$

$\therefore |A_f| > |A|$.

Loopgain $= A\beta$, if $|A\beta| = 1$, then $A_f = \infty$

$\Rightarrow$ the amplifier gives an a.c output without a.c. input signal. Hence it acts as an oscillator.

The +ve feedback

(i) increases instability of an amplifier

(ii) reduces the bandwidth

(iii) increases distortion and noise.

Negative feedback :- If ϕ_f is out of phase with ϕ_s, then

$$\boxed{\phi_i = \phi_s - \phi_f}$$

So, the i/p voltage applied to basic amplifier is decreased and correspondingly o/p is decreased. Hence, voltage gain decreased. This type of feedback is known as negative or degenerative feedback.

Gain of the amplifier with negative feedback is

$$A_f = \frac{\phi_o}{\phi_s} = \frac{\phi_o}{\phi_i + \phi_f} = \frac{1}{\frac{\phi_i}{\phi_o} + \frac{\phi_f}{\phi_o}} = \frac{1}{\frac{1}{A} + \beta}$$

$$= \frac{A}{1 + A\beta}$$

$\therefore |A_f| < |A|$, If $|A\beta| >> 1$, then $A_f \approx \frac{1}{\beta}$.

⇒ Gain depends on feedback N/w.

⇒ Negative feedback is used to improve the performance of electronic amplifiers.

⇒ It increases the bandwidth, decrease distortion and noise, modify i/p and o/p resistances as desired.

Effects of Negative feedback:-

(i) Stabilisation of gain:-

$$A_f = \frac{A}{1+A\beta}$$

$$\frac{dA_f}{dA} = \frac{1}{(1+A\beta)^2} = \frac{A_f}{A}\left(\frac{1}{1+A\beta}\right)$$

$$\frac{dA_f}{A_f} = \frac{dA}{A}\left(\frac{1}{1+A\beta}\right)$$

$\frac{dA_f}{A_f} \Rightarrow$ fractional change in amplifier voltage gain with FB

$\frac{dA}{A} \Rightarrow$ " " " without FB

$$\therefore \text{Sensitivity} = \frac{\left(\frac{dA_f}{A_f}\right)}{\left(\frac{dA}{A}\right)} = \frac{1}{1+A\beta}$$

$(1+A\beta) \Rightarrow$ desensitivity.

(ii) Increased Bandwidth:-

$BW = f_2 - f_1$, f_2 = upper cut-off frequency

f_1 = lower cut-off frequency

The product of voltage gain and BW of an amplifier without feedback and with feedback remains same.

$$\boxed{A_f \times BW_f = A \times BW}$$

As the voltage gain of FBA reduces by the factor $\frac{1}{1+A\beta}$, its BW is increased by $(1+A\beta)$. $\boxed{BW_f = (1+A\beta) \times BW}$

where A is the midband gain without FB

Due to negative feedback in the amplifier,
the upper cut off frequency f_{2f} increases and lower cut off frequency.

$$\boxed{\begin{aligned} f_{2f} &= f_2(1+A\beta) \\ f_{1f} &= \frac{f_1}{1+A\beta} \end{aligned}}$$

(iii) **Decreased Distortion**

$$\boxed{D_f = \frac{D}{1+A\beta}}$$

where D = Total Harmonic distortion

(iv) **Decreased Noise** :- N = Noise in an amplifier

$$\boxed{N_f = \frac{N}{1+A\beta}}$$

(v) **Increased input impedance** :- Ex:- (Voltage series FBA)

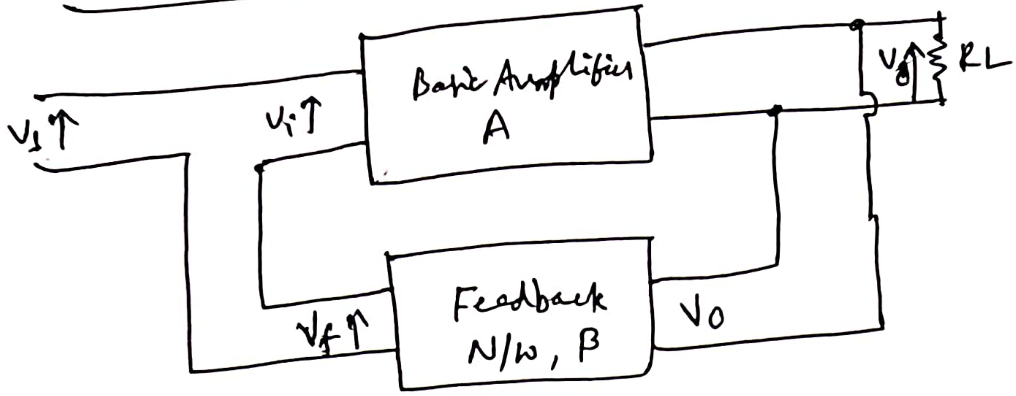
$$\boxed{Z_{if} = Z_i(1+A\beta)}$$

(vi) **Decrease in o/p Impedance** :- An amplifier with low output impedance is capable of delivering power to the load with out much loss. Ex:- (Voltage series FBA)

$$\boxed{Z_{of} = \frac{Z_o}{1+A\beta}}$$

Types of Negative feedback :-

(1) Voltage series FB :-

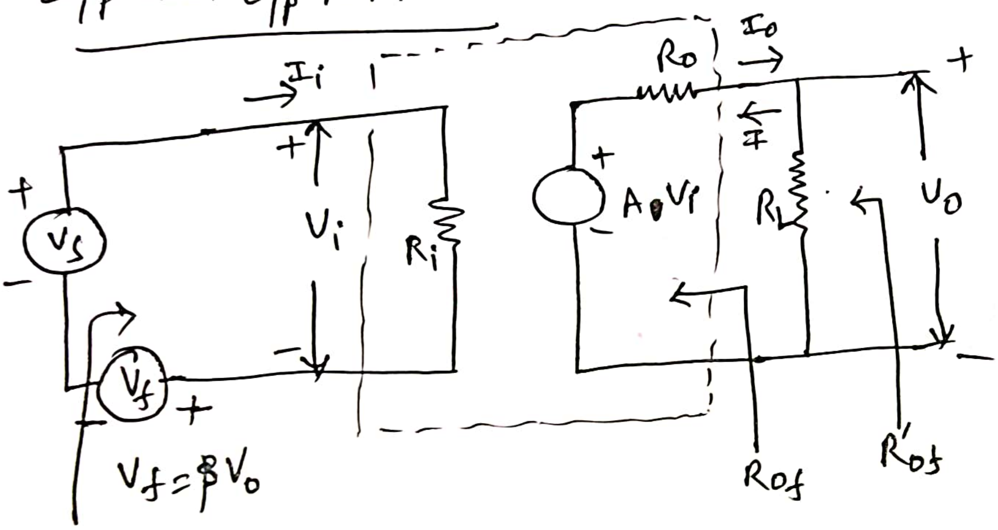


$R_{if} = R_i(1+A\beta)$

$R_{of} = \frac{R_o}{(1+A\beta)}$

$V_f \propto V_o$

I/p and O/p Resistances



$V_f = \beta V_o$

$A = \frac{V_o}{V_i}$

$\Rightarrow V_o = AV_i$

$R_{if} = V_s / I_i$

From the circuit, $V_s = V_i + V_f = I_i R_i + \beta V_o = I_i R_i + \beta A I_i R_i \quad [\because V_o = A I_i R_i]$

$$R_{if} = \frac{V_s}{I_i} = \frac{I_i R_i + \beta A I_i R_i}{I_i} = R_i(1+A\beta)$$

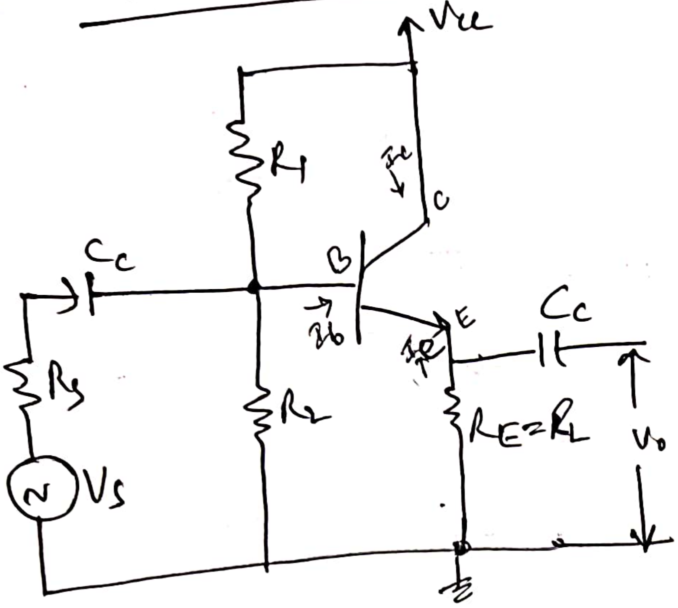
$$\boxed{\therefore R_{if} = R_i(1+A\beta)}$$

For measuring Ro, R_L is disconnected as V_s set to zero. Then an external voltage V is applied across output terminals and the current I is delivered by V is calculated. Then $R_{of} = \frac{V}{I}$.

Due to feedback, input voltage V_f reduces o/p voltage AV_i which opposes V.

$$\therefore I = \frac{V - AV_i}{R_o} = \frac{V + \beta AV}{R_o} = \frac{V(1+A\beta)}{R_o} \qquad \boxed{\therefore R_{of} = \frac{V}{I} = \frac{R_o}{1+A\beta}}$$

Emitter follower :-



The Common collector (or) Emitter follower is an example of voltage series feedback since the voltage developed in the o/p is in series with the i/p voltage as far as base-emitter junction is concerned. This is a single stage RC coupled amplifier without emitter bypass capacitor across RE. R_1 and R_2 provide base bias. The Emitter follower exhibits 100% negative feedback since the voltage at emitter follows i/p voltage.

As the o/p voltage is taken across $R_E = R_L$, feedback ratio $\beta = \frac{R_E}{R_L} = 1$

$\therefore$ overall voltage gain $A_f = \frac{A}{1+A}$, which is little less than unity.

The emitter follower increases i/p resistance and decreases o/p resistance

Characteristics of an emitter follower:

$$A_i = \frac{I_e}{I_b} = \frac{I_b + I_c}{I_b} = \frac{I_b + h_{fe} I_b}{I_b} = (1 + h_{fe})$$

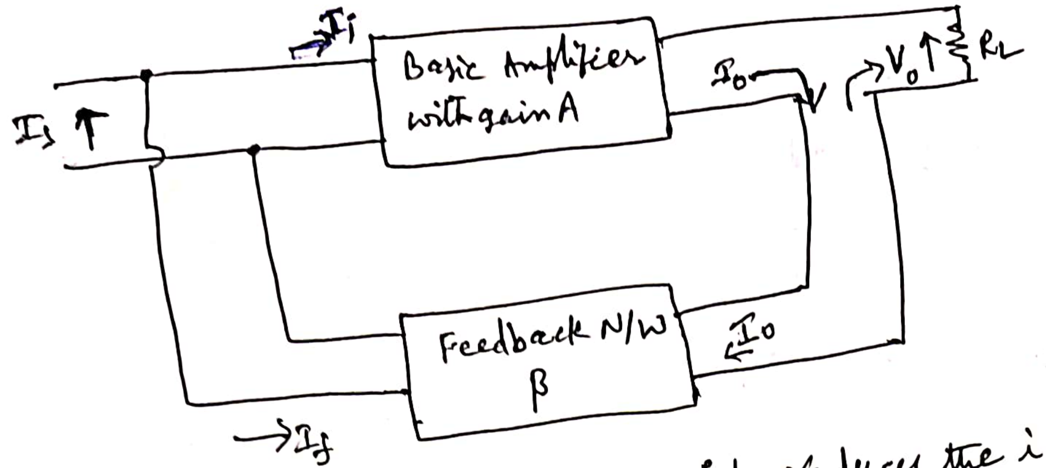
$$R_i = h_{ie} + (1 + h_{fe}) R_E = h_{ie} + (1 + h_{fe}) R_L$$

$$A_v = \frac{A_i R_L}{R_i} = \frac{(1 + h_{fe}) R_L}{h_{ie} + (1 + h_{fe}) R_L}$$

$$R_o = \frac{h_{ie} + R_s}{1 + h_{fe}}$$

$$R_{of} = R_o \,||\, R_L$$

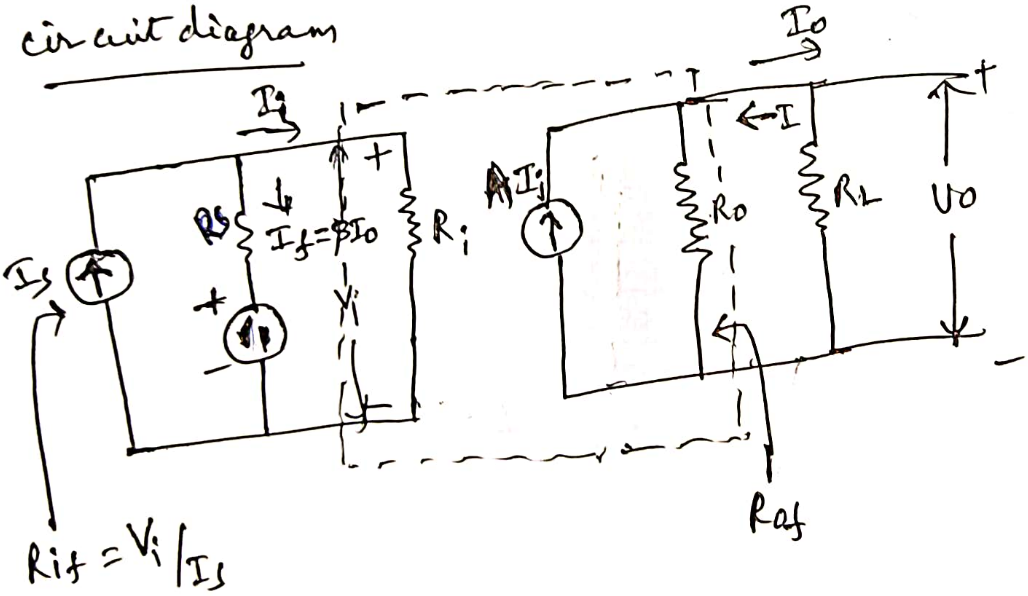
Current Shunt Feedback



$\boxed{I_f \propto I_o}$

The shunt connection at the i/p reduces the i/p resistance and the series connection at the o/p resistance increases the o/p resistance. This is a true current amplifier. The current feedback factor $\beta = \frac{I_f}{I_o}$

circuit diagram



From the ckt.

$I_s = I_i + I_f = I_i + \beta I_o$

$I_o = A I_i$

$\boxed{\therefore I_s = I_i(1 + A\beta)}$

$$R_{if} = \frac{V_i}{I_s} = \frac{V_i}{I_i(1+A\beta)}$$

$$= \frac{R_i}{1+A\beta}$$

$$\boxed{\therefore R_{if} = \frac{R_i}{1+A\beta}}$$

For measuring o/p Resistance, R_L is disconnected and V_s is set to zero. The external voltage V is applied across o/p terminals and current I delivered by V is calculated.

Then $R_{of} = V/I$

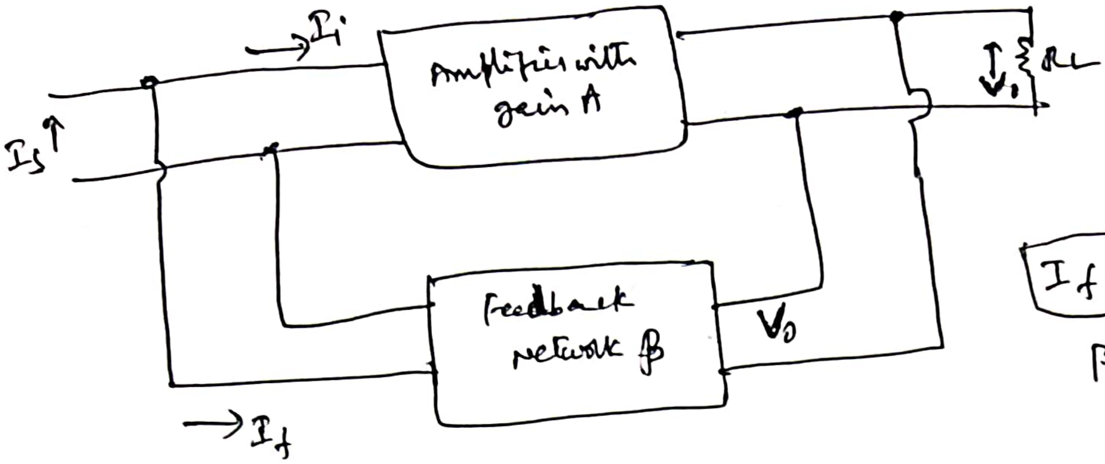
with $I_s = 0$, $I_i = -I_f = -\beta I_o = \beta I$

But, $I = \frac{V}{R_o} - \beta A I \Rightarrow I(1 + A\beta) = \frac{V}{R_o} \Rightarrow \frac{V}{I} = R_o(1 + A\beta)$

$= R_{of}$

$$\boxed{\therefore R_{of} = R_o(1 + A\beta)}$$

Voltage Shunt Feedback



Shunt derived, shunt FB connection.

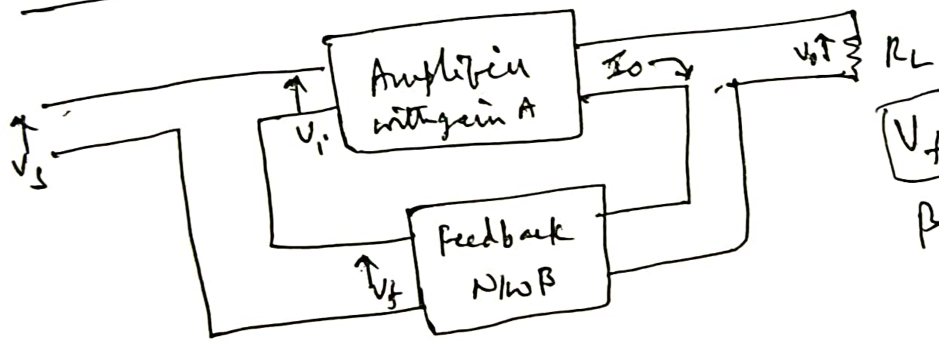
$\boxed{I_f \propto V_o}$.

$\beta = \frac{I_f}{V_o}$.

Trans resistance amplifier

$R_{if} = \frac{R_i}{1+A\beta}$, $R_{of} = \frac{R_o}{1+A\beta}$

Current Series Feedback (Transconductance amplifier)



$R_{if} = R_i(1+A\beta)$

$R_{of} = R_o(1+A\beta)$

$\boxed{V_f \propto I_o}$

$\beta = \frac{V_f}{I_o}$ (Transconductance feedback factor)

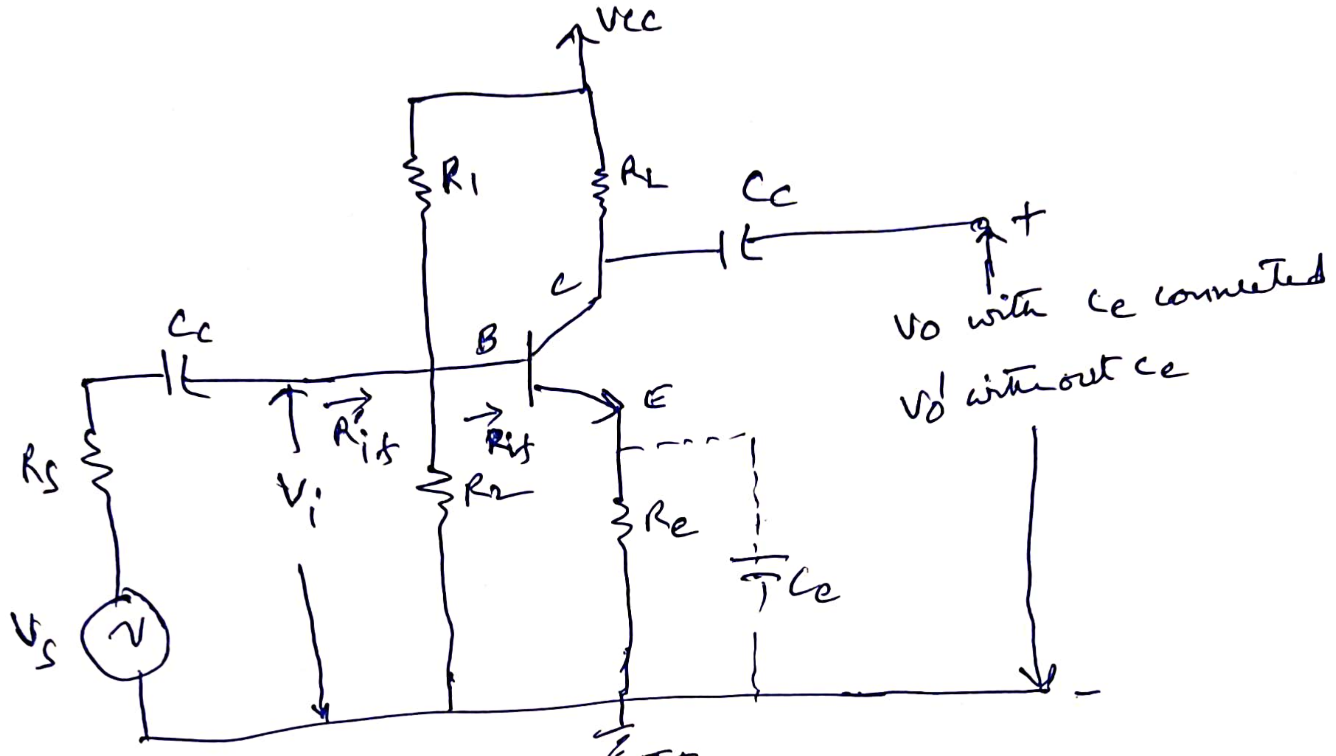
Because of series connection at i/p and o/p, the i/p & o/p resistances increases.

Effect of '-ve' feedback on Amplifier characteristics :-

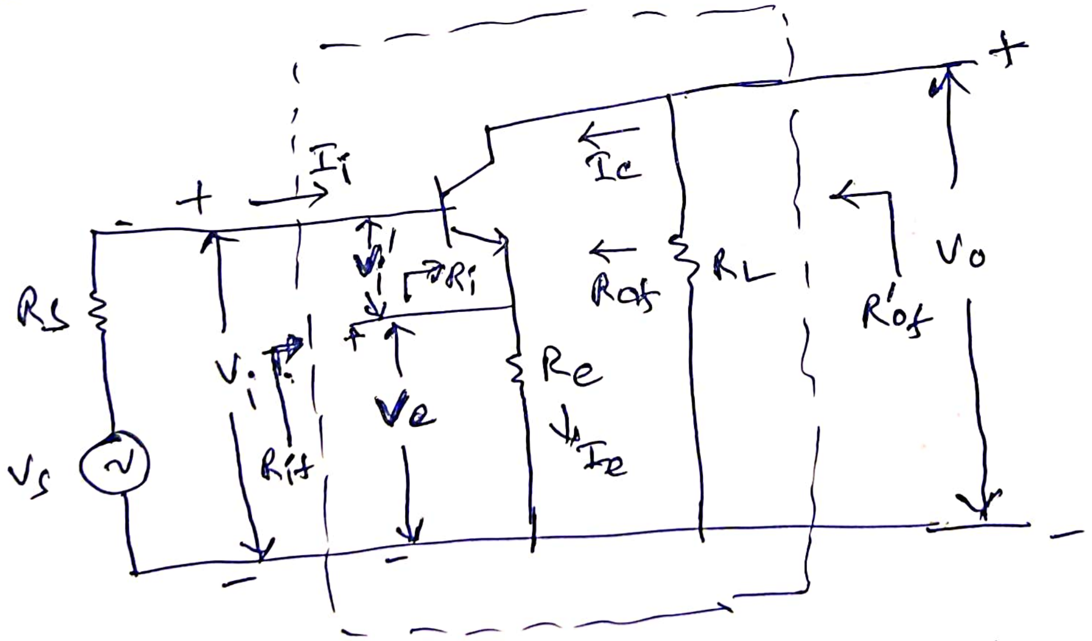
Characteristic	Current series	Voltage series	Voltage shunt	Current shunt
Voltage gain	Decreases	Decreases	Decreases	Decreases
Bandwidth	Increases	Increases	Increases	Increases
Harmonic Distortion	Decreases	Decreases	Decreases	Decreases
Noise	Decreases	Decreases	Decreases	Decreases
R_i	Increases	Increases	Decreases	Decreases
R_o	Increases	Decreases	Decreases	Increases
Other name	Transconductance Amp.	Voltage Amp	Transresistance Amp.	Current Amp

Current Series FBA

Ex:- CE Amplifier with R_e when C_e removed



Simplified diagram for ac quantities



* when R_e is properly bypassed with large C_e, output voltage is V_o and voltage gain without feedback is A.
* Resistor R_e provides dc bias stabilisation, but no ac feedback.

When C_e is removed, an a.c. voltage will be developed across R_e due to the emitter current flowing through R_e and this current is approximately equal to I_c. This voltage drop across R_e will serve to decrease the i/p voltage b/w base and emitter, so that o/p voltage will decrease to V_o'. The gain of the Amplifier with −ve feedback is A_f.

Feedback Ratio β:-

$$V_o' = I_c R_L$$

$$V_e = I_e R_e = (I_c + I_b) R_e \simeq I_c R_e$$

$$\therefore \beta = \frac{V_e}{V_o'} = \frac{I_c R_e}{I_c R_L} = \frac{R_e}{R_L}$$

Input resistance (R_{if}):- Eqn for R_{if} and A_f, with an unbypassed R_e can be calculated using fig(2)

$$R_{if} = \frac{V_i}{I_i} = \frac{V_i' + V_e}{I_i} = \frac{V_i'}{I_i} + \frac{V_e}{I_i}$$

But $R_i = \frac{V_i'}{I_i}$, also $V_e = I_e R_e$ and $I_i = I_e - I_c$

$$\therefore R_{if} = R_i + \frac{i_e R_e}{i_e - i_c} = R_i + \frac{R_e}{1 - \frac{i_c}{i_e}}$$

But, $\frac{i_c}{i_e} = h_{fb} = \frac{h_{fe}}{(1+h_{fe})}$

$$\therefore R_{if} = R_i + \frac{R_e}{1 - \frac{h_{fe}}{(1+h_{fe})}} = R_i + (1+h_{fe}) R_e$$

But $R_i = h_{ie}$

$$\boxed{\therefore R_{if} = h_{ie} + (1+h_{fe}) R_e}$$

If the bias resistance $R = R_1 || R_2 = \frac{R_1 R_2}{R_1 + R_2}$ is considered,

then $\boxed{R_{if}' = R_{if} || R}$

Voltage gain (A_f)

$$A_f = \frac{A_i R_L}{R_{if}} \quad \text{and} \quad A_i = -h_{fe}$$

$$\therefore A_f = \frac{-h_{fe} R_L}{h_{ie} + (1 + h_{fe}) R_e}$$

(Gain with feedback)

$$\text{Voltage gain without feedback} = \frac{-h_{fe} R_L}{h_{ie}} = A$$

$\therefore$ There is large decrease in voltage gain due to -ve feedback

output resistance (R_{of}):-

$$R_{of} = \frac{1 + h_{fe}}{h_{oe}} = \frac{1}{h_{ob}}$$ (approximately in MΩ)